From Raw Experience to Skill Consumption: A Systematic Study of Model-Generated Agent Skills

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arXiv 2605.23899v1 · SkillLens · independent Reel interpretation

HarnessRSI 65 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

Single-step skill feedback, not weight training

1 Collect the target's own experience
Training-side tasks; multiple rounds
Success + failure traces, outcomes, rewards

2 Extract and consolidate patterns
Up to 3 patterns per trace; merge groups of 10
Drop vague patterns at extractor discretion

3 Retain one procedural skill
≤ 3000 characters; schema + length checks
No downstream-score acceptance gate reported

4 Consume on held-out tasks
Same target; skill in the system prompt
Compare with the same target without a skill

Meta-skill changes the extractor prompt
Encode why failure happens
Give executable, domain-specific remedies
Blacklist high-risk actions

Experience/test 1:1; official train or fixed-seed split. Prompts ask to remove task details; no overlap audit reported.

2 A better skill, not a universal win

+1.55 pp vs Original · 9/9 conditions improve

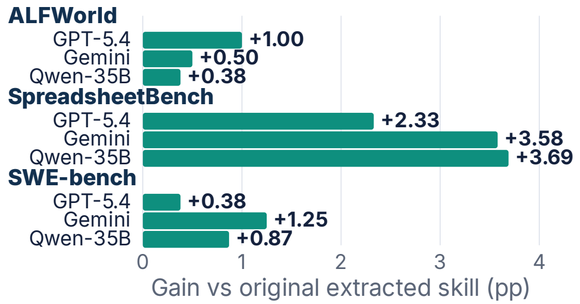


Table 11: run/seed aggregation & extractor unspecified; no CIs. Gemini/Table 1 alignment unresolved.

4 Where benefit is absent or small

No skill → Original skill → Validated skill

Table 11	No skill	Original	Validated
ALF · Qwen-35B	57.21	53.73	**54.11**
Sheet · Gemini	37.50	33.17	**36.75**

- Both validated skills still trail **no skill**.
- SWE Gemini: plausibility **70.96** exceeds validated **70.58** (Table 11).
- Table 1 TE: SWE Qwen-9B **+0.37**; SEAL Gemini-Pro **−1.04**. No cell CIs.

3 Controls and their boundaries

Matrix (Table 1): 5 domains × 6 targets × 5 extractors. Authors report 75% positive / 25% negative; ALFWorld 47% negative.

Who benefits? Spreadsheet TE: GPT-5.4 **+9.66** vs Qwen-9B **+1.46**. ALFWorld TE: GPT-5.4 **+4.93** vs Flash-Lite **−1.59**.

Repeated evaluation: Table 1, including Base, explicitly averages 3 runs. No cell SD/CI; Appendix C quantifies round noise.

Fixed-skill transfer: two Spreadsheet skills, 6 consumers. Strong-pool gain **+1.8 to +9.5**; weak-pool skill gives GPT-5.4 **−2.0** (§5.3).

Format control: four semantic-preserving rewrites, all **p > 0.34**. Table 8's model roster differs from Table 1.

Judge: 46.4% → 73.8% on the same 151 utility-labelled pairs. Rubric discovered/screened with utility; no independent rubric-design/test split described.

Access: no-skill target gets no skill; Original and both guided variants get one. No library retrieval.

Other harnesses (Table 10): CC/Codex mean **+0.4 pp**; TE **−0.5 / +2.3 / +0.7 / −1.0**.

Plausibility guidance: averages **−0.59 pp** (Table 11).

Budget: defaults shared; selection split, baseline tuning budgets and collection/extraction/rubric cost unspecified. No compute-matched no-skill control.

Takeaway

Useful skills encode concrete failure mechanisms and remedies—not just plausible advice. But utility depends on experience, extractor and consumer: even improved skills can remain worse than no skill. Keep downstream behavior as the acceptance test.

Our read · HarnessRSI

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Reports: held-out tasks, Table 1: 3-run means, controls, transfer.
Not reported: Table 11 runs/seeds & extractor; cell CIs, cost, overlap audit, rubric holdout.